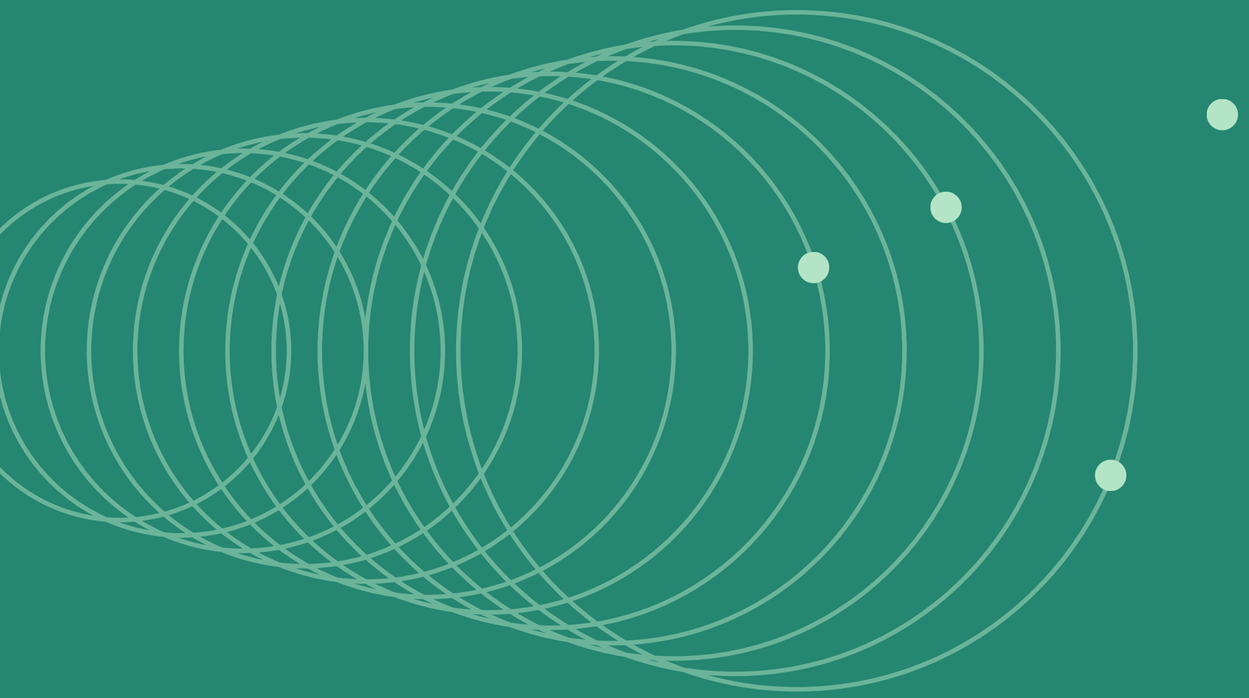
Gender Pay Gap Data Analysis

Presented by: Trinh, Amber

Agenda



Dataset Background & Description

Data Preparation & Summary

Data Visualization

Regression

Classification

References

Dataset Background & Description

1000 Employees | 9 Variables

- Brief Summary of Dataset
- Motivation
- Description of Relevant Variables

Relevant Variables

Numerical	Categorical
Age	JobTitle
PerfEval	Gender
Seniority	Education
BasePay	Dept
Bonus	

Data Preparation: Cleaning

Before	After
• 9 Variables • Basic Information	• 11 Variables ◦ TotalPay ◦ BelowMedianPay • Column Mutation ◦ Addition of relevant variables • Factor Conversion ◦ Education

Summary Measures:

Numeric Variables

Age	Min. 1st Qu. Median Mean 3rd Qu. Max. 18.00 29.00 41.00 41.39 54.25 65.00
PerfEval	Min. 1st Qu. Median Mean 3rd Qu. Max. 1.000 2.000 3.000 3.037 4.000 5.000
Seniority	Min. 1st Qu. Median Mean 3rd Qu. Max. 1.000 2.000 3.000 2.971 4.000 5.000
BasePay	Min. 1st Qu. Median Mean 3rd Qu. Max. 34208 76850 93328 94473 111558 179726
Bonus	Min. 1st Qu. Median Mean 3rd Qu. Max. 1703 4850 6507 6467 8026 11293
TotalPay	Min. 1st Qu. Median Mean 3rd Qu. Max. 40828 83443 100047 100940 117656 184010

Numeric Variable Summary by Gender

Age

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
18.00	30.00	42.00	41.83	54.00	65.00

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
18.00	28.00	40.00	41.01	55.00	65.00

PerfEval

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	3.000	2.936	4.000	5.000

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	3.000	3.126	4.000	5.000

Seniority

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	3.000	3.013	4.000	5.000

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.000	2.000	3.000	2.934	4.000	5.000

BasePay

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
34208	73186	89914	89943	106923	160614

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
36642	81452	98223	98458	115606	179726

Bonus

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1884	4841	6553	6474	8090	10768

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1703	4853	6480	6461	7989	11293

TotalPay

Female

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
40828	80866	96571	96417	112660	168968

Male

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
41030	87792	105100	104919	121617	184010

Summary Measures: Categorical Variables

Number of Employees in Each Position

Data Scientist	Driver
107	91
Financial Analyst	Graphic Designer
107	98
IT	Manager
96	90
Marketing Associate	Sales Associate
118	94
Software Engineer	Warehouse Associate
109	90

Gender Breakdown of Company

Female	Male
468	532

Highest Education of Employees

High School	College	Masters	PhD
265	241	256	238

Number of Employees in Each Department

Administration	Engineering	Management	Operations	Sales
193	192	198	210	207

Number of Employees Paid Below Median

No	Yes
500	500

Summary Measures: Categorical Variables

Gender Breakdown of Job Titles

	Female	Male
Data Scientist	53	54
Driver	46	45
Financial Analyst	49	58
Graphic Designer	48	50
IT	50	46
Manager	18	72
Marketing Associate	107	11
Sales Associate	43	51
Software Engineer	8	101
Warehouse Associate	46	44

Gender Breakdown of Departments

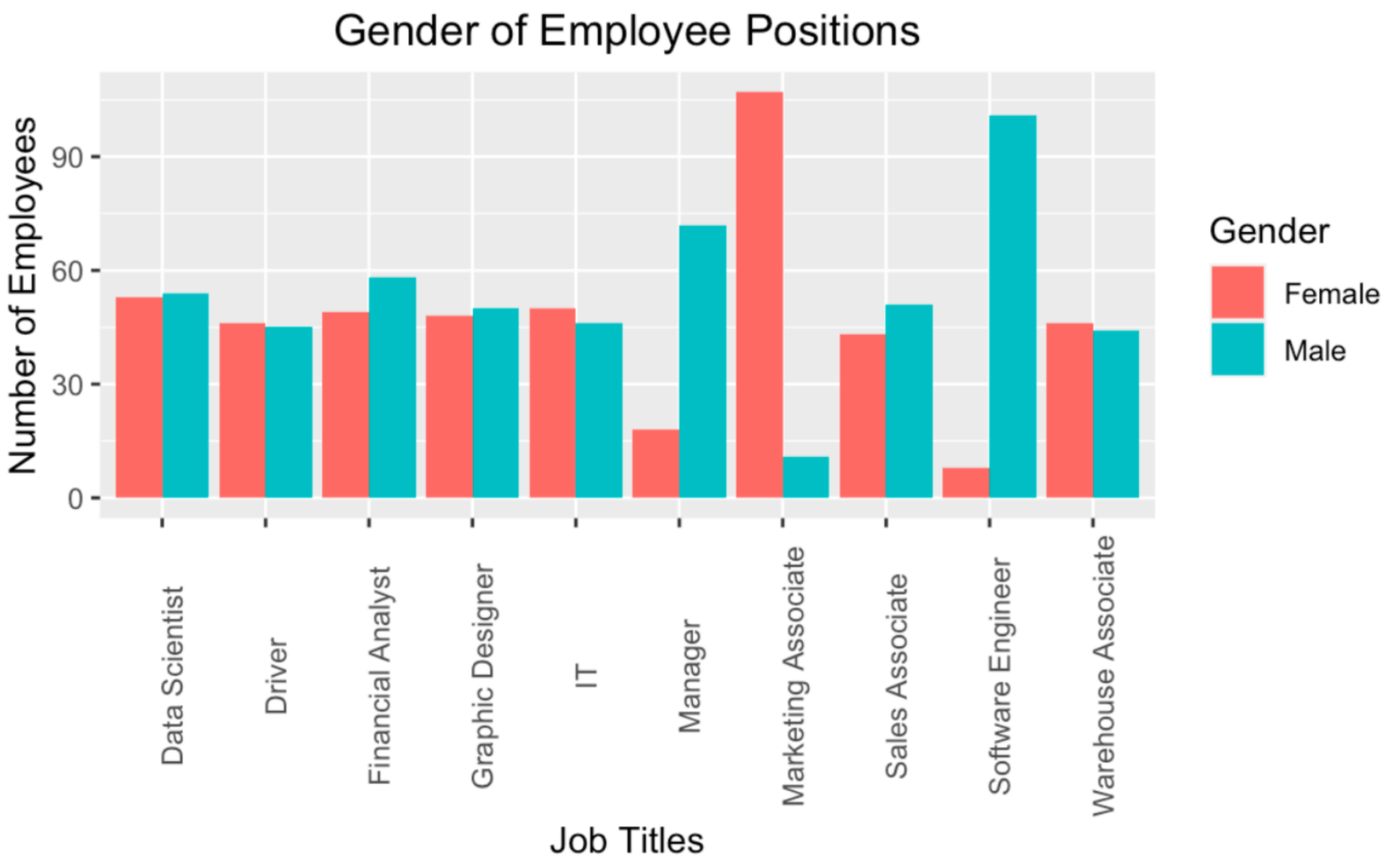
	Female	Male
Administration	95	98
Engineering	89	103
Management	87	111
Operations	96	114
Sales	101	106

Gender Breakdown of Highest Educations

	Female	Male
High School	132	133
College	123	118
Masters	107	149
PhD	106	132

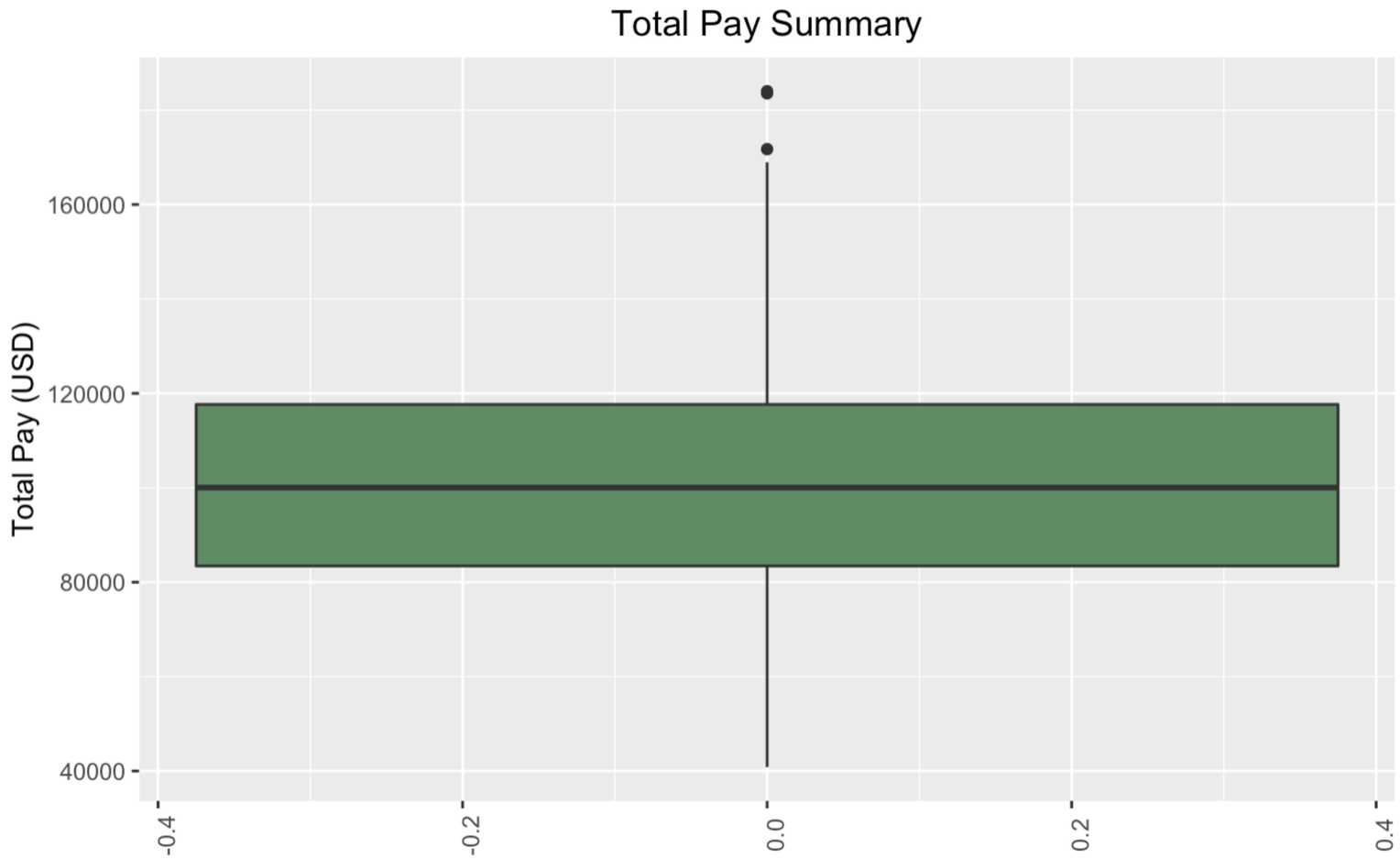
Data Visualization

How many employees take up each position in the company, and how many are male and female?



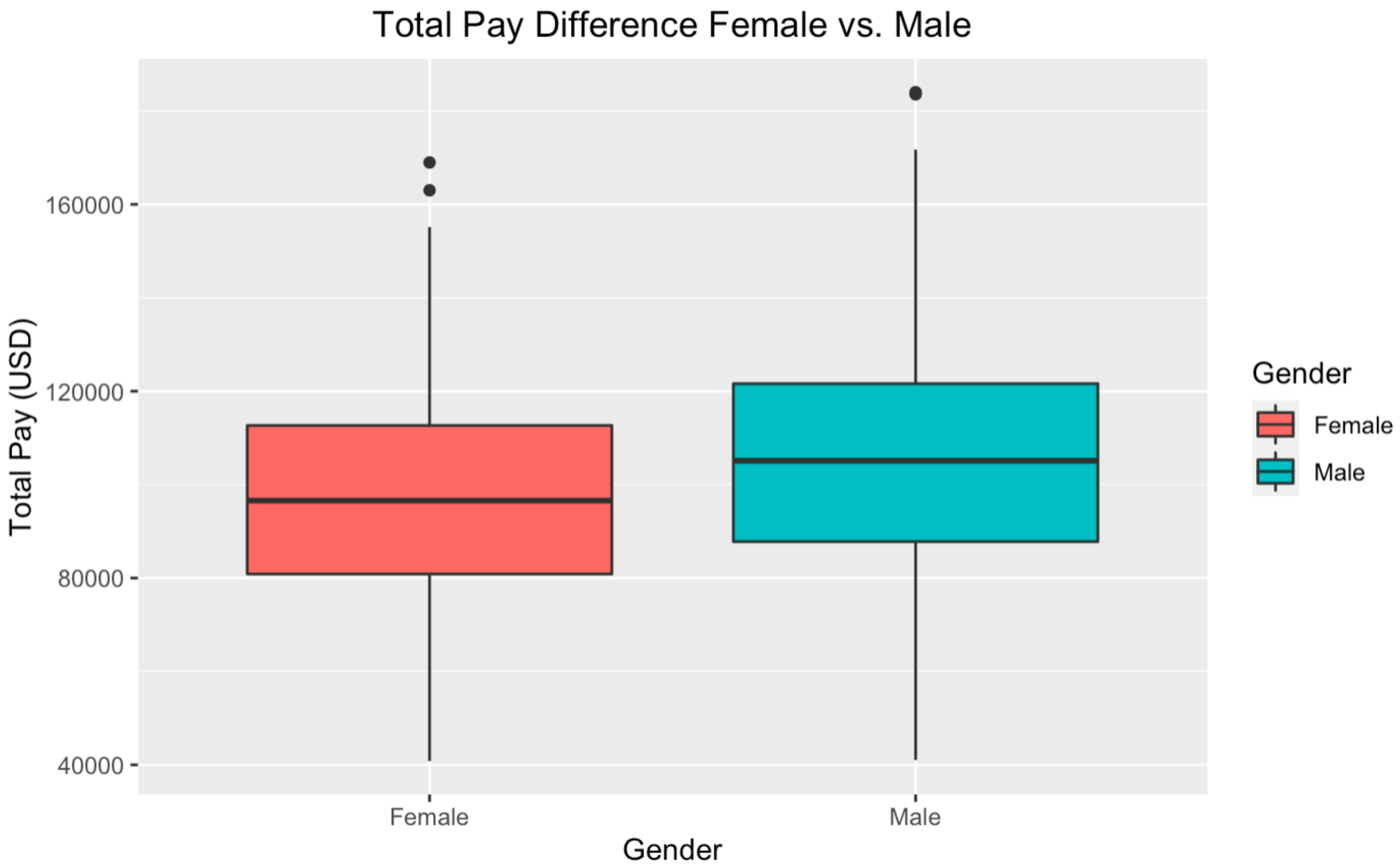
Data Visualization

What is the range of total pay across the company?



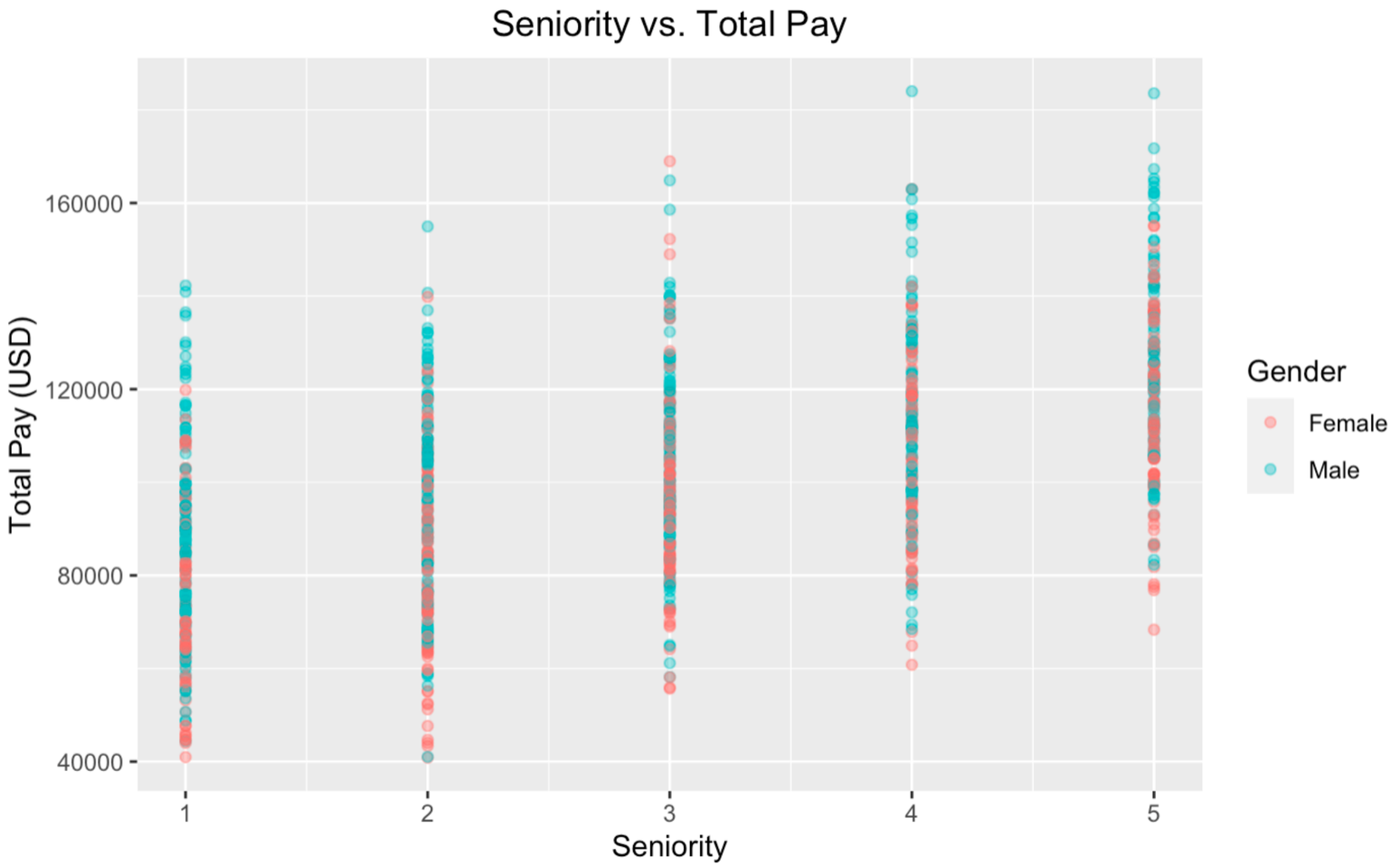
Data Visualization

What is the total pay difference between females and males?



Data Visualization

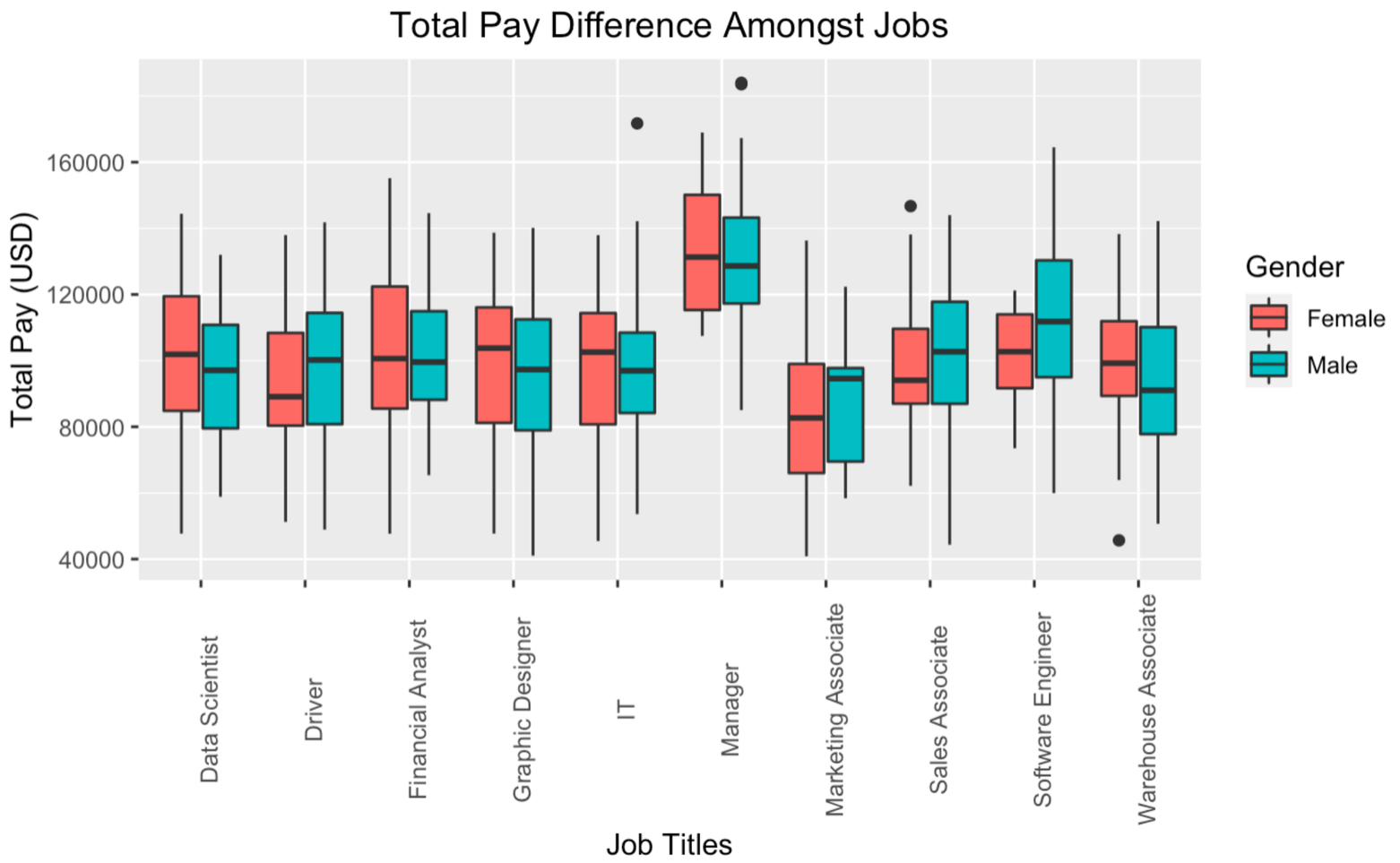
What is the relationship between Seniority level and Total Pay amongst genders?



Data Visualization

What is the difference in Total Pay across each Job Title between Genders?

Do males or females fill higher-paying jobs in the company?



Data Visualization

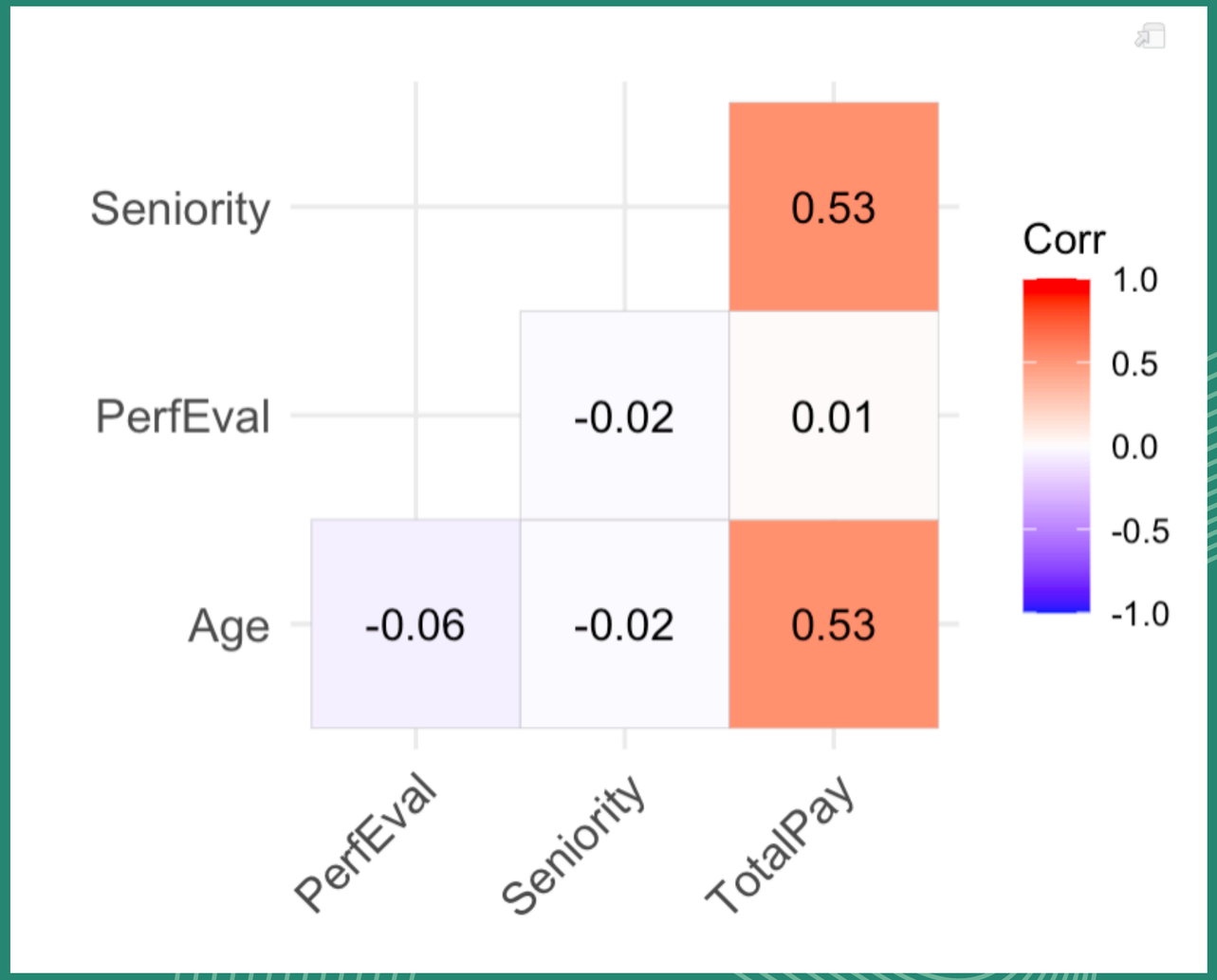
What are the top-paying jobs for each gender?

Which gender occupies the highest-paying positions in the company?



Multiple Regression

- Predicting Total Pay
  - TotalPay = 28823.78 + 965.41*Age + 1004.04*PerfEval + 9796.53*Seniority
- Making a Prediction:
  - Age = 22
  - PerfEval = 3
  - Seniority = 1
- Prediction of Total Pay:
  - $62,871.48



Age	PerfEval	Seniority
1.003760	1.003748	1.000960

Multiple Regression

- Predicting Total Pay
  - $\hat{y}$ = 28823.78 + 965.41 (Age) + 1004.04 (PerfEval) + 9796.53 (Seniority)
- Making a Prediction:
  - Age = 22
  - PerfEval = 3
  - Seniority = 1
- Prediction of Total Pay:
  - $62,871.48

```
Call:
lm(formula = .outcome ~ Age + PerfEval + Seniority, data = dat)

Residuals:
   Min     1Q Median     3Q    Max
-46981 -11041  -1126   9164  59002

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 28823.78    2289.22  12.591  < 2e-16 ***
Age           965.41      36.13  26.718  < 2e-16 ***
PerfEval     1004.04     362.74   2.768  0.00575 **
Seniority    9796.53     369.74  26.495  < 2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

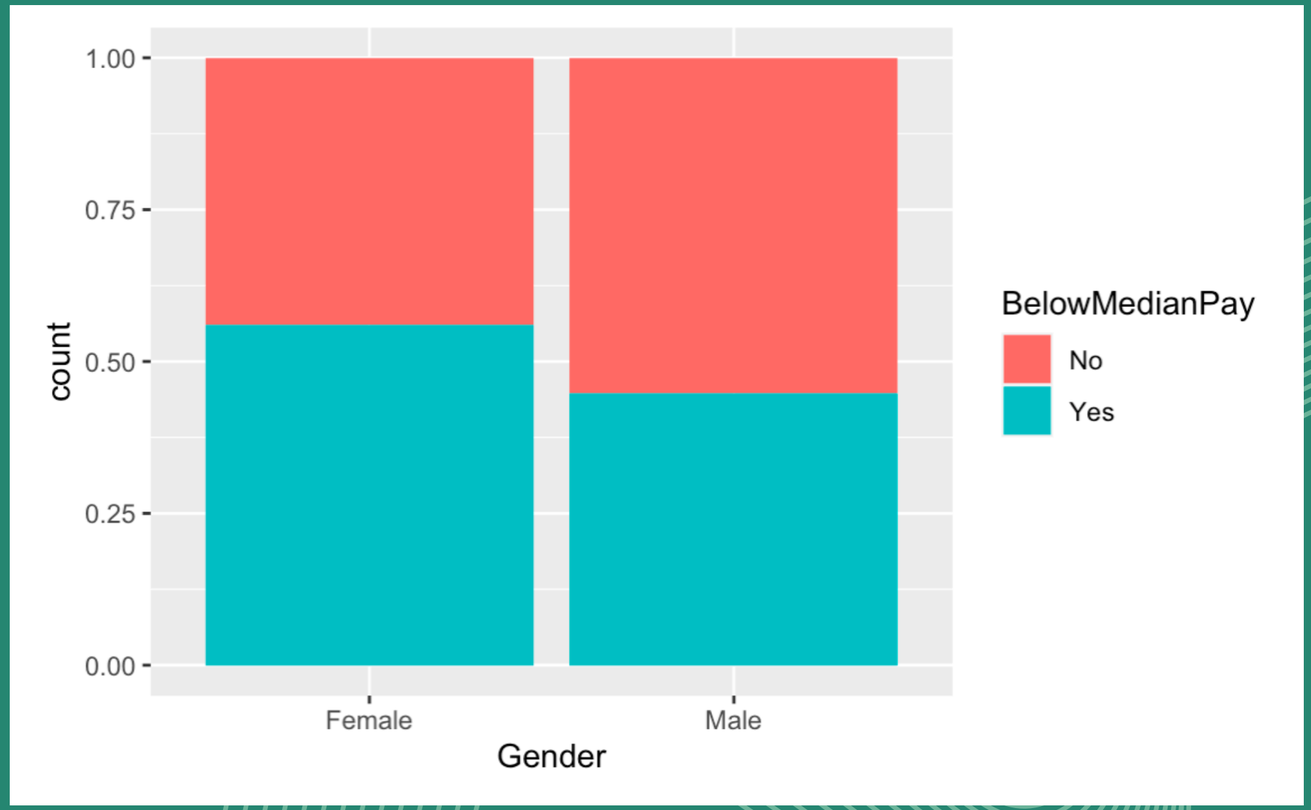
Residual standard error: 16300 on 996 degrees of freedom
Multiple R-squared:  0.5817,    Adjusted R-squared:  0.5804
F-statistic: 461.7 on 3 and 996 DF,  p-value: < 2.2e-16
```

Logistic Regression

- Predicting Below Median Pay
  - ln(p/(1-p)) = 9.80726 -0.12221*Age - 0.13662*PerfEval - 1.26649*Seniority

- Making a Prediction:
  - Age = 22
  - PerfEval = 3
  - Seniority = 1

- Prediction of Below Median Pay:
  - Yes

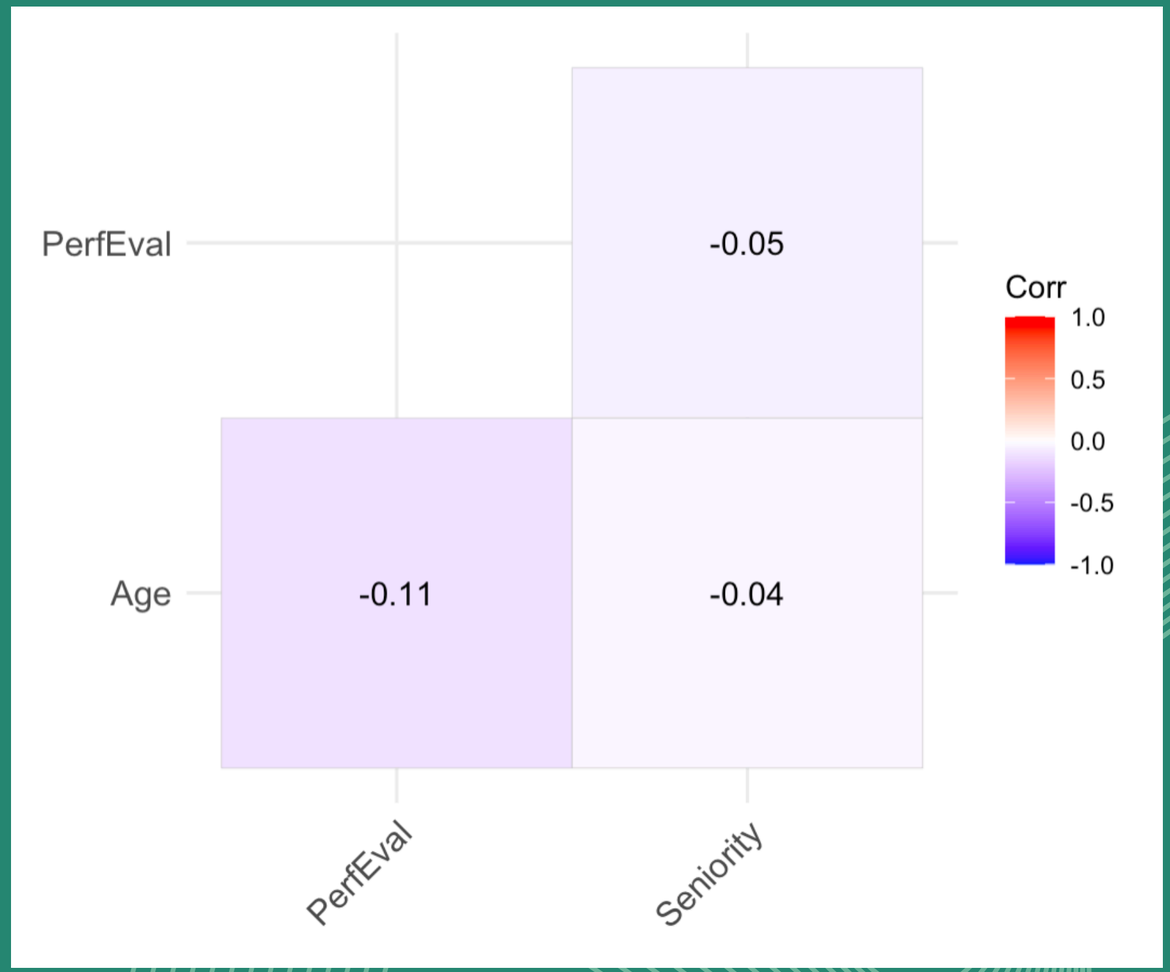


Logistic Regression

- Predicting Below Median Pay
  - ln(p/(1-p)) = 9.80726 -0.12221*Age - 0.13662*PerfEval - 1.26649*Seniority

- Making a Prediction:
  - Age = 22
  - PerfEval = 3
  - Seniority = 1

- Prediction of Below Median Pay:
  - Yes



Age	PerfEval	Seniority
1.631388	1.048160	1.594313

Logistic Regression

- Predicting Below Median Pay
  - ln(p/(1-p)) = 9.80726 -0.12221*Age - 0.13662*PerfEval - 1.26649*Seniority

- Making a Prediction:
  - Age = 22
  - PerfEval = 3
  - Seniority = 1

- Prediction of Below Median Pay:
  - Yes

```
Call:
NULL

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.8164  -0.5961   0.1541   0.6206   2.4867

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  9.80726    0.95524  10.267   <2e-16 ***
Age         -0.12221    0.01268  -9.636   <2e-16 ***
PerfEval    -0.13662    0.09163  -1.491    0.136
Seniority   -1.26649    0.12709  -9.965   <2e-16 ***
---
Signif. codes:
0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 642.07  on 467  degrees of freedom
Residual deviance: 380.21  on 464  degrees of freedom
AIC: 388.21

Number of Fisher Scoring iterations: 5
```

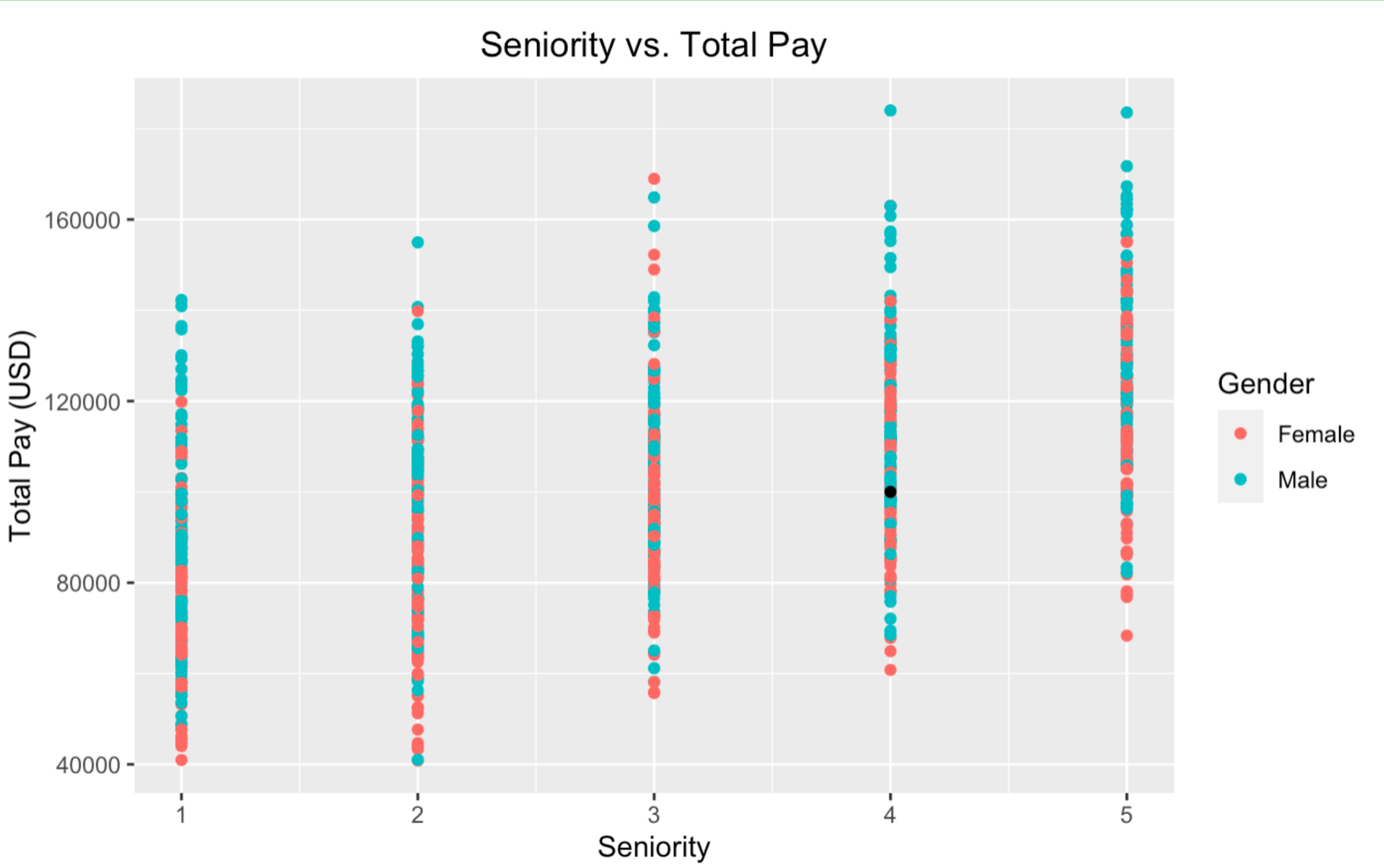
```
 (Intercept)          Age     PerfEval    Seniority
1.816520e+04 8.849649e-01 8.722995e-01 2.818193e-01
```

Classification

K-Nearest Neighbor

Classifying an employee's gender

- Seniority = 4
- Total Pay = 100,000



Classification

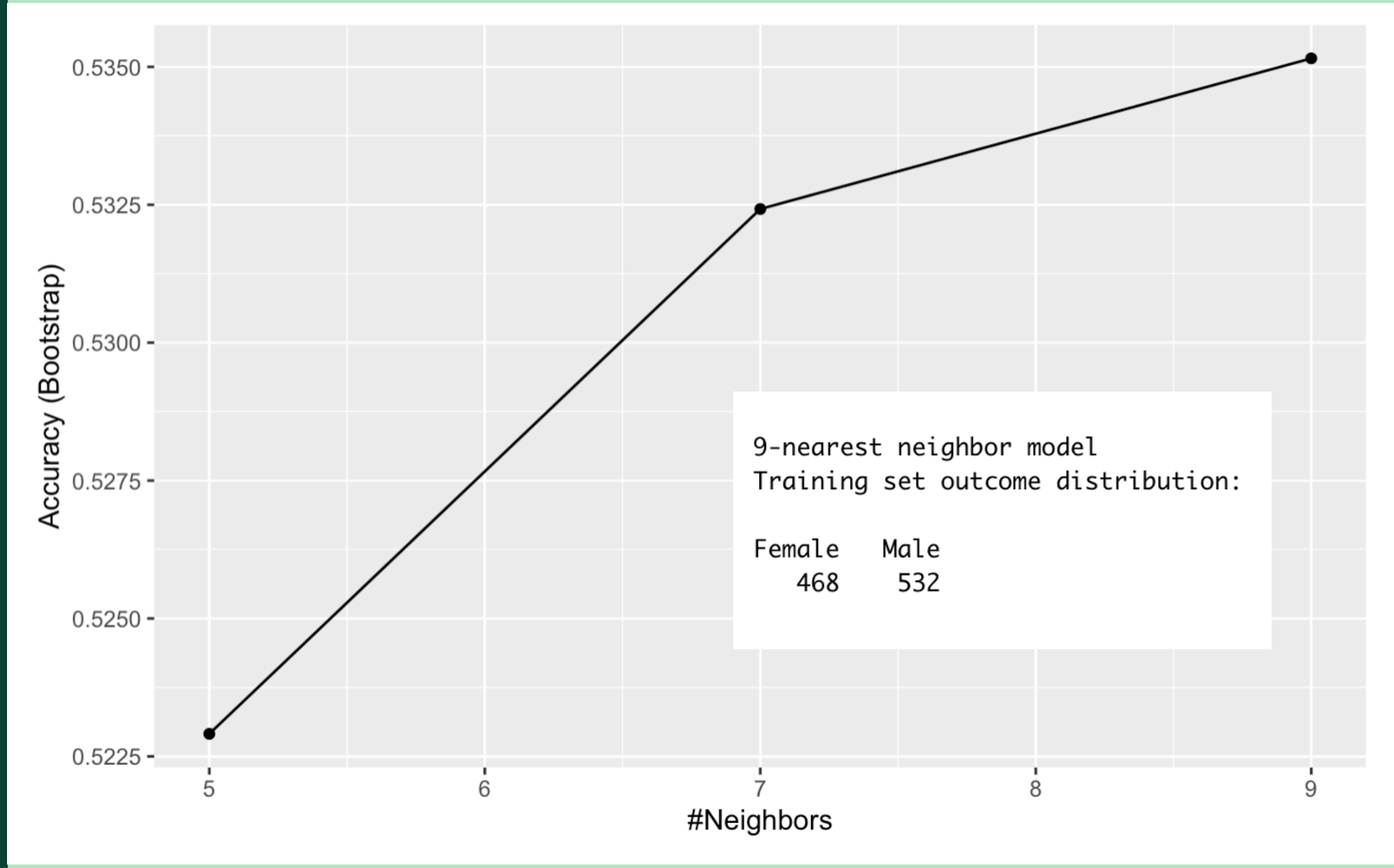
K-Nearest Neighbor

Making a Prediction:

- Seniority = 4
- TotalPay = 100000

Prediction:

- Male



References

- http://faculty.cas.usf.edu/mbrannick/regression/Logistic.html
- https://www.glassdoor.com/research/how-to-analyze-gender-pay-gap-employers-guide/
- https://r-graph-gallery.com/264-control-ggplot2-boxplot-colors.html
- http://sape.inf.usi.ch/quick-reference/ggplot2/colour
- https://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704-ep713_multivariablemethods/bs704-ep713_multivariablemethods2.html